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9

NUMERICAL METHODS FOR TRANSITION REGIME FLOWS

9.1 Classification of methods

As was noted in the preceding chapter, very few of the 'analytical' methods lead to closed form solutions and most depend on numerical procedures for the production of the final results. On the other hand, the 'numerical' methods rely on computation from the outset. While the analytical methods are almost invariably based on the Boltzmann equation which is the accepted mathematical model of a gas flow at the molecular level, a large fraction of the numerical methods are based directly on the simulation of the physics of the flow and do not depend on a mathematical model. This difference in approach can be used to define two classes of methods, but the situation is complicated by the efforts that have been made to link some of the simulation methods to the Boltzmann equation with sufficient rigour that they can be claimed to be 'solutions' of that equation. All methods that employ simulated particles or molecules will be classed here as simulation methods.

A general difficulty with simulation methods is that they are not unique and there may be many variants of a particular method. Furthermore, modifications to a method, or to the computational procedures through which it is implemented, may alter the relative advantages, or even the validity of that method. Successive applications of a method by the same person may use quite different procedures, even though the description of the method is not changed. Conversely, changes to the procedures that are well within the usual variations between implementations may be claimed to define a 'new' method which is given a different name. This can lead to confusion and to difficulties with 'quality control'. While all numerical methods are subject to these problems, physical simulation methods suffer from the fact that they are not as clearly defined as the conventional numerical methods for the solution of equations. The problem has been compounded by the restrictions on the length and content of scientific papers, in that these do not allow a full exposition of the theory behind the procedures, let alone the actual coding of the computer programs. Also, as the simulation becomes more exact through the inclusion of additional physical effects, the programs become more substantial and a large number of test cases are required for code verification prior to 'production runs' being made.

9.2 Direct Boltzmann CFD

The most direct numerical approach to the Boltzmann equation would be to apply the conventional methods of computational fluid dynamics (or CFD). The velocity distribution function is the only dependent variable for a simple gas and this may be represented in a finite difference or finite element form by its numerical value over a network of points or elements in phase space.

A major problem is posed by the sheer number of points or elements that must be employed for this representation. For a steady one-dimensional problem with gradients in only one spatial dimension, the distribution function is axially symmetric in velocity space and a three-dimensional array is required in phase space. In addition, the velocity space domain is of infinite extent. Finite bounds must therefore be chosen such that only a negligible fraction of molecules lie outside them, but it is difficult to predict the proper placement of these bounds. This problem is particularly severe in the case of hypersonic or high-temperature flows because a minute fraction of molecules with very high velocities relative to the bulk of the gas can have a significant effect on the macroscopic flow properties. The velocity distribution function becomes three-dimensional for two-dimensional flows, thus requiring a five-dimensional array in phase space. If the flow is unsteady, time must be added as a further independent variable. If 100 grid intervals were employed for each dimension, the finite difference or element solution of the Boltzmann equation for a three-dimensional unsteady flow would require 10^{14} grid points. Suitably coupled simultaneous equations must be solved for each of the molecular species in a gas mixture. Also, each of the internal modes that are associated with these species add further to the effective dimensions of phase space. Physical effects such as chemical reactions and thermal radiation pose almost insurmountable problems for the Boltzmann formulation, if only because of the difficulty in defining inverse collisions.

In view of these difficulties, it is no surprise that direct solutions of the Boltzmann equation have been restricted to simple flow geometries, monatomic gases, and low to moderate Mach numbers. Even so, there are problems due to the very large number of operations that are required to evaluate the collision term. The integrals are replaced by summations in the discrete approximation. At each point in phase space, a sum must be taken over all other velocity space points with each term in the sum representing, itself, a summation over the impact parameters of the collision. The direct solutions have employed summation procedures that reduce the computational effort associated with the collision term.

The first successful method for the direct numerical solution of the Boltzmann equation was introduced in a series of papers by Nordsieck, Hicks, and Yen (for example: Nordsieck and Hicks 1967, and Yen 1970). They dealt with one-dimensional steady flow problems and employed a conventional finite difference technique for the 'fluid-like' terms on the left-hand side of the equation. A Monte Carlo sampling technique was used to

reduce the computing effort required by the summations associated with the collision term. The steady flow equation was employed and this necessitated an initial estimate of the distribution function over the whole flowfield, with the final solution being approached through an iterative procedure. The number of points in the computational grid was minimal and, since collisions were represented by transitions between points in the velocity space grid, there was an effective rounding off of the velocities in collisions. This led to a systematic error which tended to build up from step to step in the iteration. A least squares procedure was used to keep the error in check by ensuring that the conservation laws were satisfied. In addition to these new problems associated with the collision term, the finite difference method for the remaining terms posed the usual problems with regard to grid dependence and stability. Despite these difficulties, satisfactory results were obtained for a number of problems; most notably the structure of normal shock waves (Hicks et al. 1972).

The Boltzmann CFD approach was extended to two-dimensional steady flows by Tcheremissine (1973). Alternatives to the Monte Carlo procedure for the evaluation of the collision integral have been proposed and tested by Tcheremissine (1991) and Tan et al. (1989). These methods attempt to reduce the computational load associated with the collision term by concentrating on the more non-equilibrium portion of the velocity distribution function. The basic disadvantage of the Boltzmann CFD methods is the necessity to specify a bounded grid in velocity space. This causes problems with regard to the overall computational load and with the ease of use of the method. Any attempt to ease the computational load by applying location dependent procedures to velocity space tends to make the method even more difficult to apply.

9.3 Deterministic simulations

While the particulate nature of a gas is responsible for the mathematical difficulties that are associated with the Boltzmann equation, it also permits the circumvention of these problems through the development of physically based simulation methods. Direct simulation methods model the real gas by a large number of simulated molecules in a computer. This number has ranged from the order of hundreds in early calculations, to millions in some contemporary simulations. The position coordinates, velocity components, and internal state of each of the simulated molecules are stored in the computer and are modified with time as the molecules are simultaneously followed through representative collisions and boundary interactions in simulated physical space. The time parameter in the simulation may be identified with physical time in the real flow and all calculations are unsteady, although steady flow may be attained as the large-time state of the unsteady flow. Because a steady flow develops through a physically real unsteady flow, there are minimal problems with boundary conditions. There is no need for an initial estimate of the flowfield and there is no artificial iteration process.

The *molecular dynamics (or MD) method* (Alder and Wainwright 1957) was the first physical simulation method. While this method may employ probabilistic procedures when setting the initial configuration of the molecules, the calculation of the subsequent molecular motion, including the collisions and boundary interactions, is deterministic. Collisions occur when the cross-sections overlap, and the computation time for a straightforward application of the method is proportional to the square of the number of simulated molecules. A major difficulty with this method is that, for a given molecular size, flow geometry, and gas density, the number of simulated molecules is not a free parameter.

The parameter that best expresses the number of molecules that are involved in a flow is the number of molecules within one cubic mean free path. For a nominal molecular diameter of 4×10^{-10} m, eqn (1.58) gives this number as

$$N_\lambda \equiv n\lambda^3 = 3856(n_0/n)^2, \tag{9.1}$$

where n_0 is the standard number density. If we consider the MD simulation of a problem that requires a three-dimensional flow field that extends over 30 mean free paths in each direction, approximately 10^8 simulated molecules would be required at standard density. However, for the same calculation at a density of the order of 100 times standard density, the required number is reduced to the easily attained 10^4. MD calculations with a realistic molecular diameter are therefore restricted to dense gases, and the overwhelming number of MD applications fall into this category. Note that the Boltzmann equation is restricted to dilute gases, and this requires the density to be no greater than about twice the standard density.

At the other extreme, problems in rarefied gas dynamics typically involve a density ratio of the order of 10^{-6}, and eqn (9.1) then shows that the required number of simulated molecules would be 10^{20}. If the MD method is to be applied to low-density problems, the correct mean free path can be obtained only through the device of replacing the large number of small molecules by a small number of large molecules. This leads to errors due to an incorrect equation of state, biased impact parameters and, in some cases, the clustering or trapping of molecules in acute corners and small openings. The best-known such application is that of Meiburg (1986) to the two-dimensional unsteady flow past an inclined flat plate with a chord length of 75 mean free paths. The required molecular diameter was 0.75 times the mean free path, or 1% of the chord of the plate. MD calculations must be three-dimensional and Meiburg's calculation employed a thin 'slice' with specularly reflecting boundaries. This procedure would not be possible for axially symmetric flows which would appear to require a full three-dimensional calculation. It may be noted that many two-dimensional MD calculations employ cylindrical rather than spherical molecules, and this leads to non-physical macroscopic properties. The MD method has proved to be valuable for the simulation of dense gases and liquids (Allen and Tildesley 1987), but it is inappropriate for dilute gases.

9.4 Probabilistic simulation methods

The first probabilistic simulation method was introduced by Haviland and Lavin (1962) and is best termed the *test particle Monte Carlo method*. The test particle method for collisionless flows was presented in §7.7 as the logical method for dealing with complex flows that involve multiple surface reflections. In the transition regime, it becomes necessary to compute typical intermolecular collisions in addition to the molecule-surface interactions. This can only been done if there is already a representation of the complete flowfield. The distribution function is chosen for this representation and, as in the finite difference method, it is stored at a number of locations in phase space. The method therefore shares a major disadvantage of the finite difference method in that an initial estimate must be made of the distribution function over the whole flowfield. A large number of 'test particle' trajectories are computed with the assumed distribution serving as the 'target' gas for the computation of typical intermolecular collisions. An updated target distribution is then constructed from the history of the test or incident molecules. The process is continued until there is no difference between the target and incident distributions.

The alternative to the test particle approach is to introduce a time variable and to follow the trajectories of a very large number of simulated molecules simultaneously within a computer. The first paragraph in the preceding section applies equally well to this method, called the *direct simulation Monte Carlo (or DSMC) method*, but the probabilistic procedures depend on the dilute gas assumption. Therefore, while the MD method is strictly applicable only to dense gas flows, the DSMC method can be applied only to dilute gas flows. The DSMC method was first applied to the homogeneous gas relaxation problem (Bird 1963), and the first application to a flow was to the shock structure problem (Bird 1965). Hammersley and Handscomb (1964) state that 'Monte Carlo methods comprise that branch of experimental mathematics which is concerned with experiments on random numbers'. They also applied the term 'direct simulation' to the modelling of probabilistic problems, which is the simplest form of the Monte Carlo method. The term 'direct simulation Monte Carlo' is therefore a generic term that more than covers the many variants of the method that have been introduced over the years.

The essential DSMC approximation is the uncoupling, over a small time interval or 'step', of the molecular motion and the intermolecular collisions. All the molecules are moved (including the computation of the resulting boundary interactions) over the distances appropriate to this time step, followed by the calculation of a representative set of intermolecular collisions that are appropriate to this interval. The time step should be small in comparison with the mean collision time and, as long as this condition is met, experience has shown that the results are independent of its actual value. The molecule movement routines are trivial, and a simple indexing of the molecules to sub-cells as well as cells causes all collisions to be between near neighbours. The procedures for the probabilistic

selection of a representative set of collisions are based directly on the relations that have formed the basis of kinetic theory for more than a century. Because of this, the method has the same limitations as classical kinetic theory, which includes the Boltzmann equation. The principal limitations are the assumption of molecular chaos and the requirement of a dilute gas. The DSMC method cannot be applied to dense gases or to highly ionized plasmas that are dominated by long-range interactions.

The cell or spatial element network is required only in physical space, with the velocity space information being contained in the positions and velocities of the simulated molecules. The physical space network is used to facilitate the choice of the molecules for collisions and for the sampling of the macroscopic flow properties. Advantage may be taken of symmetries in physical space to reduce the number of dimensions of the cell network, and also the number of position coordinates that need to be stored for each molecule, but the collisions are always treated as three-dimensional phenomena. Note that the positions and velocities of the simulated molecules are continuously distributed, velocity space is unbounded, and the conservation of mass, momentum, and energy is imposed to within the round-off error of the computer.

The microscopic boundary conditions are specified by the behaviour of the individual molecules, rather than in terms of the distribution function. This facilitates the incorporation of complex physical effects, such as chemical reactions, into the simulation. The DSMC method employs simulated molecules of the correct physical size and their number is reduced to a manageable level by regarding each simulated molecule as representing a fixed number F_N of real molecules. This is typically a very large number and advantage can sometimes be taken of its size to specify some properties through a distribution of values instead of a single value for each simulated molecule. The macroscopic boundary conditions are usually specified as a uniform gas at zero time. The flow develops from this initial state with time in a physically realistic manner, rather than by iteration from an initial approximation to the flow. All procedures may be specified such that the computational time is directly proportional to the number of simulated molecules. The computational efficiency of the DSMC method is far higher than that of the MD method.

The DSMC method can be applied to weakly ionized gases as long as the electric field is obtained from theoretical considerations or calculations that are external to the DSMC procedures. The field cannot be obtained directly from the simulation because the spurious fields due to statistical fluctuations in the electron and ion number densities would generally be large in comparison with the real fields. The problem may be avoided when the Debye distance is very small through the enforcement of charge neutrality by moving the electrons with the ions. Finite electric fields can be calculated by a simultaneous solution of the Poisson equation, as in the study by Sugimura and Vogenitz (1975). Alternatively, the field in a shock front may be estimated (Bird 1986) from an approximate theory for ambipolar diffusion.

9.5 Discretization methods

The *lattice gas* cellular automata (Boon 1990) and similar methods have a computational efficiency that is even higher than that of the DSMC method, but this is achieved at the expense of physical reality. In fact, these methods are more accurately described as providing gas analogues, rather than gas models.

The lattice gas takes its name from the regular array of points or nodes into which physical space is discretized. Time is also discretized to uniform steps, the particle speeds are restricted either to one or to just a few values, and their directions are restricted to those of the lines that join adjacent nodes. The symmetry requirements for a fluid simulation require that a plane or two-dimensional lattice should be triangular rather than square. The information on the state of each node is contained in a word with a length of the order of one byte. The velocities and time step are such that the particles move exactly to another node, generally to a neighbouring node. The interactions between the nodes are performed through operations on the individual bits of these words. Rules for the interactions have been developed such that the continuity and momentum equations are satisfied and the system as a whole exhibits a fluid-like behaviour. However, there are difficulties in establishing the temperature, the energy equation is not properly defined, and the gas properties do not resemble those of any real gas. A quantitative comparison of the results from the triangular lattice model[†] with experimental data and analytical results has been made by Long et al. (1987).

Nadiga et al (1989) investigated a nine-velocity model based on a square lattice that included the diagonals. There were three speeds such that the particles were either stationary or moved to an adjacent node at each time step. They were able to define a Knudsen number, showed that the Boltzmann H function indicated an approach to equilibrium, and demonstrated the production of shock waves. Also, while the system was found to be reversible, it was very sensitive to random perturbations when moving in the reverse direction. However, the system exhibited a number of unphysical features, and it was not possible to define the thermodynamics with sufficient precision to allow the behaviour of the model to be used to predict that of a real gas.

The use of a plane lattice means that the *gas* as well as the *flow* is two-dimensional. Three-dimensional lattices have been developed, and these require a word length of the order of three bytes to describe each node. The fact that vacant as well as occupied nodes add to the storage requirement is a disadvantage of the lattice gas model. This becomes particularly serious in the case of three-dimensional calculations and can negate the storage savings of the approximate lattice models as compared with the the more realistic DSMC molecular models.

[†] This is sometimes called a hexagonal lattice because the bisectors of the lines joining the nodes define a hexagon.

Cellular automata are a field of study in their own right and they provide insight to phenomena that involve the collective behaviour of a very large number of particles. However, when applied to gas flows, the level of physical realism is such that they are of little use for engineering studies.

The *discrete ordinate or discrete velocity method* was introduced by Broadwell (1964) and generalized by Gatignol (1970). The discretization of velocity space allows the Boltzmann equation to be replaced by a set of non-linear hyperbolic differential equations. The number of equations in this set is equal to the number of discrete velocities, and the collision term involves a double summation over the velocity points. The solution of this set is a formidable task and applications have been limited to simple geometries and simple molecular models.

The number of discrete velocities in some applications of the discrete ordinate method is so small that the method resembles an analytical form of the lattice gas model. On the other hand, some versions of the DSMC method have employed a very large number of discrete velocities (e.g. Goldstein et al. 1989). Gropengiesser et al. (1991) introduced a method called the 'Finite Pointset Method' that employs discretization in both physical and phase space. The relative advantages and disadvantages of continuously distributed and discrete molecular locations and velocities in direct simulation methods will be discussed in the following chapter.

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